

UNIT 1

SETS



SET

A **set** in Mathematics is a collection of objects that are clearly defined and well-organized. These objects can be numbers, letters, shapes, or anything else, and each object in the set is called a member or an element.

FORMS OF SET

Tabular form of a set:

The elements of a set that are listed within brackets $\{ \}$ and separated by commas are called tabular form.

Example: Set of first five small letters of the English alphabets. In tabular form we write as $\{a, b, c, d, e\}$

Descriptive form

The descriptive form of a set is a way to represent a set using words or symbols to describe its elements, we write a description to explain what is in the set.

Here are some examples:

Set A: The set of all students in Class VII of your school.

Set B: The set of the first ten natural numbers.

Set-builder Notation

Set-builder notation defines a set by describing the properties or rules that its elements must satisfy, rather than listing them. It is written as $\{x \mid \text{property of } x\}$, where x is an element of the set, and $\mid$ means such that.

Example: Set $B = \{x \mid x \text{ is an even number greater than } 2\}$ This describes the set of even numbers greater than 2.

TYPES OF A SET

1. Empty Set: It is an empty set with no elements.

Example: $\{ \}$, It is also denoted by Φ .

2. Singleton Set: It is a set that has only one element.

Example: $\{2\}$

3. Finite Set: A finite set has a countable number of elements.

Example: $\{1, 2, 3, 4, 5\}$

4. Infinite Set: An infinite set has an unlimited number of elements that continue forever.

Example: $\{1, 2, 3, 4, 5, 6, \dots\}$

SET NOTATION

1. $\{\}$ (Curly Braces): Curly braces are used to write or show a set. The elements of the set are placed inside these braces.

Example: $\{1, 2, 3\}$

2. $\in$ (Element Of): The symbol $\in$ means "is an element of" or "belongs to." It is used to show that a specific item is part of a set.

Example: $2 \in \{1, 2, 3\}$ (2 is an element of the set).

3. $\notin$ (Not Element Of): Indicates that an element does not belong to a set.

Example: $4 \notin \{1, 2, 3\}$ (4 is not an element of the set).

4. $\subseteq$ (Subset): Indicates that one set is a subset of another (all elements of the first set are also in the second set).

Example: $\{1, 2\} \subseteq \{1, 2, 3\}$

SET OPERATIONS

1. Union ($\cup$): The set of all elements that are in either or both sets.

Example: If $A = \{1, 2\}$ and $B = \{2, 3\}$, then $A \cup B = \{1, 2, 3\}$.

2. Intersection ($\cap$): The set of elements common to both sets.

Example: If $A = \{1, 2\}$ and $B = \{2, 3\}$, then $A \cap B = \{2\}$.

3. Difference ($-$): The set of elements in one set that are not in the other.

Example: If $A = \{1, 2, 3\}$ and $B = \{2, 3\}$, then $A - B = \{1\}$.

4. Complement ($'$): The set of elements not in the given set, within a universal set.

Example: If $U = \{1, 2, 3, 4\}$ and $A = \{1, 2\}$, then $A' = \{3, 4\}$.

EXERCISE 1.1

1. Which of the following are sets? Justify your answer.

- (a) The collection of all the days in a week beginning with the letter 'T'.
 (b) The collection of all beautiful flowers in the park.
 (c) The collection of girls in your class.
 (d) The collection of all rivers in Pakistan.
 (e) The collection of all active teachers in the school.
 (f) The collection of all integers more than -3.

2. If, $A = \{3, 5, 7, 9\}$, $B = \{2, 4, 6, 8, 10\}$, $C = \{12, 14, 18, 20, 24\}$ and $D = \{21, 26, 31, 36\}$

(a) State whether true or false.

i. $13 \in C$ _____

ii. $6 \notin A$ _____

iii. $9 \notin A$ _____

iv. $24 \in C$ _____

v. $36 \notin D$ _____

(b) Fill in the blanks.

(i) 5 _____ A

(ii) 4 _____ B

(iii) 26 _____ C

(iv) $8 \notin$ _____

(v) $3 \in$ _____

(ii) 6 _____ C

(iii) 21 _____ D

(iv) 18 _____ B

3. Write the following sets in the tabular form.

- (a) A = The set of all even numbers less than 12
 (b) B = The set of all prime numbers greater than 1 but less than 29
 (c) C = The set of integers lying between -2 and 2
 (d) D = The set of letters in the word LOYAL

- (e) F = The set of all factors of 36
 (f) $G = \{x \mid x \in \mathbb{N}, 5 < x < 12\}$
 (g) $H = \{x \mid x \text{ is a multiple of 3 and } x < 21\}$
 (h) $M = \{x \mid x \text{ is a positive integer and } x < 40\}$

4. Write each of the following in set descriptive form.

- | | |
|-----------------------------------|--------------------------------------|
| (a) $A = \{5, 10, 15, 20\}$ | (e) $H = \{\text{Jan, June, July}\}$ |
| (b) $B = \{1, 2, 3, 6, 9, 18\}$ | (f) $I = \{a, e, i, o, u\}$ |
| (c) $C = \{P, R, I, N, C, A, L\}$ | (g) $J = \{a, b, c, d, \dots, z\}$ |
| (d) $D = \{0\}$ | (h) $L = \{1, 3, 5, 7, 9\}$ |

5. Classify the following as finite and infinite sets.

- (a) $A = \{x \mid x \in \mathbb{N} \text{ and } x \text{ is even}\}$
 (b) $B = \{x \mid x \in \mathbb{N} \text{ and } x \text{ is composite}\}$
 (c) $C = \{x \mid x \in \mathbb{N} \text{ and } 3x - 2 = 0\}$
 (d) $D = \{x \mid x \in \mathbb{N} \text{ and } x^2 = 9\}$
 (e) $E = \{\text{The set of numbers which are multiples of 3}\}$
 (f) $G = \{\text{The set of letters in the English alphabet}\}$
 (g) $H = \{\text{The set of persons living in a house}\}$

6. Identify the following as null set or singleton set.

- (a) $A = \{x \mid x \in \mathbb{N}, 1 < x < 2\}$ _____
 (b) $P = \{\text{Point of intersection of two lines}\}$ _____
 (c) $C = \{x \mid x \text{ is an even prime number greater than 2}\}$ _____
 (d) $Q = \{x \mid x \text{ is an even prime number}\}$ _____
 (e) $E = \{x \mid x^2 = 9, x \text{ is even}\}$ _____
 (f) $R = \{\text{The set of whole numbers lying between 0 and 2}\}$ _____
 (g) $B = \{0\}$ _____
 (h) $D = \{\text{The set of largest 1 digit number}\}$ _____
 (i) $F = \{\text{The set of triangles having 4 sides}\}$ _____

Union of Sets

The union of two or more sets is a set containing all elements from each set, without duplicates.

Notation:

$A \cup B$ (Union of sets A and B)

Example 1:

If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$ find $A \cup B$.

Solution:

$$A \cup B = \{1, 2, 3, 4\} \cup \{3, 4, 5, 6\}$$

$$A \cup B = \{1, 2, 3, 4, 5, 6\}$$

Example 2:

If $A = \{a, b, c\}$ and $B = \{c, d, e\}$ find $A \cup B$.

Solution:

$$A \cup B = \{a, b, c\} \cup \{c, d, e\}$$

$$A \cup B = \{a, b, c, d, e\}$$

Explanation:

The union includes all elements from A and B. The duplicate element "c" is only included once.

Example 3:

If $A = \{1, 3, 5\}$, $B = \{2, 4, 6\}$ and $C = \{1, 2, 3\}$ find $A \cup B \cup C$.

Solution:

$$A \cup B \cup C = \{1, 3, 5\} \cup \{2, 4, 6\} \cup \{1, 2, 3\}$$

$$A \cup B \cup C = \{1, 2, 3, 4, 5, 6\}$$

Intersection of Sets

The intersection of two or more sets is a set containing only the elements common to all sets.

Notation:

$A \cap B$ (Intersection of sets A and B)

Example 1:

If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$ find $A \cap B$.

Solution:

$$A \cap B = \{1, 2, 3, 4\} \cap \{3, 4, 5, 6\}$$

$$A \cap B = \{3, 4\}$$

Example 2:

If $A = \{a, b, c\}$ and $B = \{c, d, e\}$ find $A \cap B$.

Solution:

$$A \cap B = \{a, b, c\} \cap \{c, d, e\}$$

$$A \cap B = \{c\}$$

Example 3:

If $A = \{1, 3, 5\}$, $B = \{2, 4, 6\}$ and $C = \{1, 2, 3\}$ find $A \cap B \cap C$.

Solution:

$$A \cap B \cap C = \{1, 3, 5\} \cap \{2, 4, 6\} \cap \{1, 2, 3\}$$

$$A \cap B \cap C = \emptyset \text{ (Empty Set)}$$

Set Difference

The difference of two sets, A and B, is a set containing elements in A but not in B.

Notation:

$$A - B$$

Example 1:

If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$ find $A - B$.

Solution:

$$A - B = \{1, 2, 3, 4\} - \{3, 4, 5, 6\}$$

$$A - B = \{1, 2\}$$

Example 2:

If $A = \{a, b, c\}$ and $B = \{c, d, e\}$ find $A - B$.

Solution:

$$A - B = \{a, b, c\} - \{c, d, e\}$$

$$A - B = \{a, b\}$$

Example 3:

If $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$ find $B - A$.

Solution:

$$B - A = \{4, 5, 6\} - \{1, 2, 3\}$$

$$B - A = \{4, 5, 6\}$$

EXERCISE 1.2

1. Let $A = \{x \mid x \text{ is a natural number and a factor of } 18\}$

$B = \{x \mid x \text{ is a natural number and less than } 6\}$

Find $A \cup B$ and $A \cap B$.

2. If $P = \{\text{multiples of } 3 \text{ between } 1 \text{ and } 20\}$ and $Q = \{\text{even natural numbers up to } 15\}$.

Find the intersection of the two given set P and set Q.

3. $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$.

Find the difference between the two sets:

(i) A and B

(ii) B and A

4. Let $A = \{a, b, c, d, e, f\}$ and $B = \{b, d, f, g\}$.

Find the difference between the two sets:

(i) A and B

(ii) B and A

PROPERTIES OF UNION OF SETS

The properties of the set union indicate their nature and help carry out further operations. Here are some properties of the union of sets.

Properties of Union	Notation
Commutative Property	$A \cup B = B \cup A$
Associative Property	$(A \cup B) \cup C = A \cup (B \cup C)$

Commutative Property

$$A \cup B = B \cup A$$

The union sets will be independent of the order of the sets.

Example: set $A = \{\text{orange, banana, apple, mango}\}$ and set $B = \{1, 2, 4, 6\}$.

$$A \cup B = \{\text{orange, banana, apple, mango, } 1, 2, 4, 6\}$$

$$B \cup A = \{\text{orange, banana, apple, mango, } 1, 2, 4, 6\}$$

Associative Property

$$(A \cup B) \cup C = A \cup (B \cup C)$$

The union of three or more sets does not depend on how we group the sets.

Example: $A = \{\text{orange, banana, apple, mango}\}$, $B = \{1, 2, 4, 6\}$ and $C = \{A, E, I, O, U\}$ prove $(A \cup B) \cup C = A \cup (B \cup C)$.

$$A \cup B = \{\text{orange, banana, apple, mango, 1, 2, 4, 6}\}$$

$$(A \cup B) \cup C = \{\text{orange, banana, apple, mango, 1, 2, 4, 6, A, E, I, O, U}\}$$

$$(B \cup C) = \{1, 2, 4, 6, A, E, I, O, U\}$$

$$A \cup (B \cup C) = \{\text{orange, banana, apple, mango, 1, 2, 4, 6, A, E, I, O, U}\}$$

$$\text{Thus, } (A \cup B) \cup C = A \cup (B \cup C)$$

The operation of intersection in sets also follows the commutative and associative properties. That is

$$A \cap B = B \cap A$$

$$(A \cap B) \cap C = A \cap (B \cap C)$$

Set Complement

The complement of a set A , written as A' , is the set of all elements that are not in set A but are part of the universal set.

Example 1:

Let Universal Set $(U) = \{1, 2, 3, 4, 5, 6\}$ and $A = \{1, 2, 3\}$ find A' .

Solution:

$$U - A = \{1, 2, 3, 4, 5, 6\} - \{1, 2, 3\}$$

$$A' = \{4, 5, 6\}$$

Example 2:

Let Universal Set $(U) = \{a, b, c, d, e\}$ and $A = \{a, c\}$ find A' .

Solution:

$$U - A = \{a, b, c, d, e\} - \{a, c\}$$

$$A' = \{b, d, e\}$$

EXERCISE 1.3

1. Let $A = \{0, 1, 2, 3, 4, 5\}$, $B = \{2, 4, 6, 8\}$ and $C = \{1, 3, 5, 7\}$

Verify $(A \cap B) \cap C = A \cap (B \cap C)$

2. Let A , B and C be three sets such that:

Set $A = \{2, 4, 6, 8, 10, 12\}$, set $B = \{3, 6, 9, 12, 15\}$ and set

$C = \{1, 4, 7, 10, 13, 16\}$.

Find:

(i) $A \cup B$

(iii) $B \cap A$

(ii) $A \cap B$

(vi) Is $A \cup B = B \cup A$

3. If $A = \{1, 3, 7, 9, 10\}$, $B = \{2, 5, 7, 8, 9, 10\}$, $C = \{0, 1, 3, 10\}$,
 $D = \{2, 4, 6, 8, 10\}$, $E = \{\text{negative natural numbers}\}$ and $F = \{0\}$

Find:

(i) $A \cup B$

(ii) $E \cup D$

4. If $A = \{2, 3, 4, 5\}$, $B = \{c, d, e, f\}$ and $C = \{4, 5, 6, 7\}$;

Find:

(i) $(A \cup B) \cap (A \cup C)$

(ii) $A \cup (B \cap C)$

5. If $A = \{a, b, c, d\}$, $B = \{c, d, e, f\}$ and $C = \{b, d, f, g\}$;

Find:

(i) $(A \cap B) \cup (A \cap C)$

(ii) $A \cap (B \cup C)$

(iii) Is $(A \cap B) \cup (A \cap C) = A \cap (B \cup C)$

6. Find the complement of set A and set B also show

$(A \cup B)' = A' \cap B'$, where $U = \{11, 12, 13, 14, 15, 16\}$, $A = \{12, 13\}$ and $B = \{13, 14, 15\}$.

7. Let U be the universal set which consists of all the integers greater than 5 but less than or equal to 25. Let A and B be the subsets of U defined as:

A = $\{9, 16, 25\}$

B = $\{7, 9, 16, 18, 24\}$

Find the complement of sets A and B and $A' \cap B'$.

8. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $A = \{3, 4, 5\}$, $B = \{4, 5, 6\}$. Show that $(A \cup B)' = A' \cap B'$.

9. If $U = \{a, b, c, d, e, f, g, h\}$, $P = \{a, c, d\}$, $Q = \{a, b, f, g\}$. Prove the Demorgan's first law $(P \cup Q)' = P' \cap Q'$ and De Morgan's second law $(P \cap Q)' = P' \cup Q'$

10. Given $U = \{10, 11, 12, 13, 14, \dots, 20\}$, $A = \{10, 12, 14, 16, 18\}$ and $B = \{10, 11, 12, 13, 14, 15\}$, Verify the following.

1. $U' = \emptyset$

2. $\emptyset' = U$

3. $A \cup A' = U$

4. $B \cap B' = \emptyset$

5. $(A \cup B)' = A' \cap B'$

6. $(A \cap B)' = A' \cup B'$

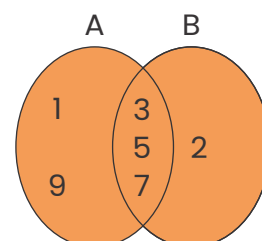
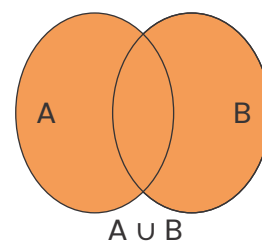
VENN DIAGRAM

Venn diagram is a visual representation or illustration that uses circles to show relationships between finite sets. Each circle represents a specific set. Any operation on sets can be represented using a Venn diagram.

Venn Diagram of Union of Sets

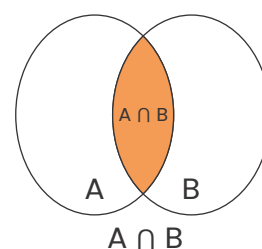
The union of sets is illustrated by the orange shaded region. It covers all the elements from both the sets. Here, U is a universal set, a big set such that A and B are subsets of U .

Example: Set $A = \{1, 3, 5, 7, 9\}$ and Set $B = \{2, 3, 5, 7\}$
 $A \cup B = \{1, 2, 3, 5, 7, 9\}$



What is $A \cap B$?

The intersection of two sets, indicated by " $A \cap B$ ", is the collection of elements shared by both sets A and B . In other words, it contains all of the elements found in both sets.



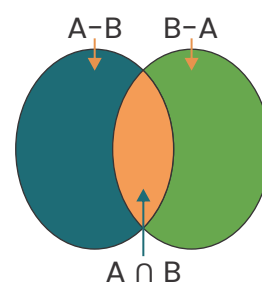
Set Difference in Venn Diagrams

A Venn diagram is the pictorial representation of sets and their different operations.

Observe the Venn diagram shown below.

Two circles that are overlapping symbolize sets A and B .

The overlapping part shows the intersection of A and B .



- The region of Set A that does not intersect with Set B represents $A - B$. The blue area shows the elements that are exclusively found in Set A and are not present in Set B.
- The region of Set B that does not intersect with Set A represents $B - A$. It is represented with green colour in the example below.

Example 1:

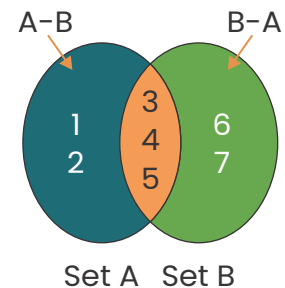
Find the set difference between Set

$A = \{1, 2, 3, 4, 5\}$ and Set $B = \{3, 4, 5, 6, 7\}$.

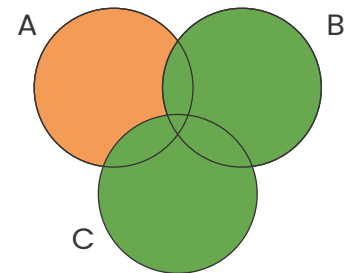
Let's draw the Venn diagram.

$$A - B = \{1, 2, 3, 4, 5\} - \{3, 4, 5, 6, 7\} = \{1, 2\}$$

$$B - A = \{3, 4, 5, 6, 7\} - \{1, 2, 3, 4, 5\} = \{6, 7\}$$

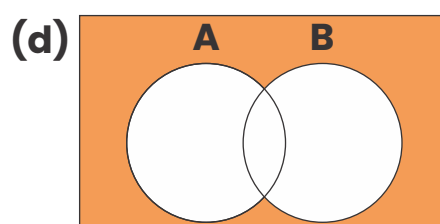
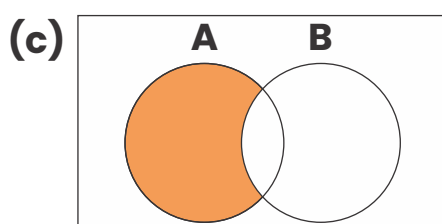
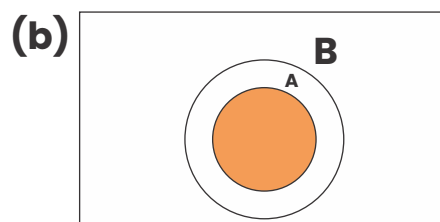
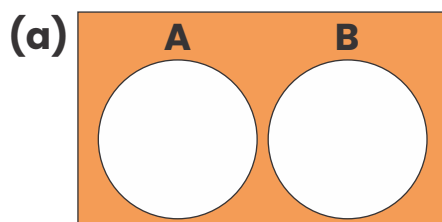
**Difference of Three Sets**

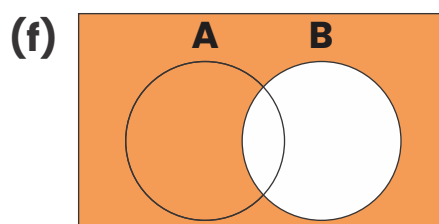
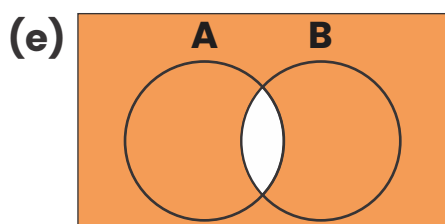
Let A, B, and C are three non-empty sets, then the difference $A - B - C$ represents the set that has the elements of A that are not in B and C.

**EXERCISE 1.4****1. Draw Venn diagrams to represent**

(a) Union of two sets

(b) Intersection of two sets

2. What set is represented by the shaded portion in the following Venn diagrams?



3. Let $A = \{1, 3, 5, 7\}$ and $B = \{2, 4, 6, 8\}$ find the $A \cup B$ and $A \cap B$ and represent operations through Venn Diagram.

Facts about Union of Sets

De Morgan's Law:

The complement of the union of two sets is the same as the intersection of the complements of the individual sets.

Formula: $(A \cup B)' = A' \cap B'$

Size of the Union:

The union of two sets is always larger than or equal to the size of the individual sets.

Example: If $A = \{1, 2\}$ and $B = \{3, 4\}$, then $A \cup B = \{1, 2, 3, 4\}$, which is larger than each set.

Union with the Empty Set:

The union of a set with the empty set ($\emptyset$) is the original set.

Formula: $A \cup \emptyset = A$

Example: If $A = \{1, 2, 3\}$, then $A \cup \emptyset = \{1, 2, 3\}$.